KARTIK KARKERA

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Education

University of Texas, Dallas **May 2026**
Master of Science in Computer Science *GPA: 3.67 / 4.0*

Technical Skills

Languages: Python, SQL
Big Data & Processing: Apache Spark (PySpark, Spark SQL), Delta Lake, Databricks, Kafka
Orchestration & ELT: Apache Airflow, ETL/ELT Pipelines
Cloud & Infrastructure: Amazon Web Services, Docker, Kubernetes
Databases & Warehousing: PostgreSQL, DynamoDB, Elasticsearch, MongoDB

Certifications

Databricks Certified Associate Developer for Apache Spark - Python Jan 2026
AWS Certified Solutions Architect – Associate (SAA-C03) Jul 2025
AWS Certified Data Engineer – Associate (DEA-C01) Sept 2025

Experience

PyCray Technologies Noida, India
Software Engineer *May 2023 - Jun 2024*

- Engineered an event-driven ETL pipeline for an algorithmic trading platform using DynamoDB Streams and AppSync subscriptions, processing real-time position updates with ess than 50ms end-to-end latency.
- Designed a hot-cold data architecture separating active trading data (DynamoDB) from historical analytics (PostgreSQL), decoupling operational and analytical query paths to maintain low-latency reads without degrading reporting workloads.
- Built scheduled EventBridge-triggered Lambda ETL jobs to migrate trading records from DynamoDB to PostgreSQL, implementing schema evolution logic to maintain data consistency across pipeline versions.
- Optimized location-based search on a carpooling platform by designing geospatial indexing strategies in Elasticsearch, reducing query latency by 50% while maintaining data consistency under concurrent writes.
- Developed an analytics aggregation pipeline over trading and user datasets to compute KPIs including active users, open positions, and PnL trends, enabling daily internal monitoring and business reporting.
- Built a data aggregation layer using Hasura GraphQL to transform raw transaction logs into structured analytical metrics, powering internal BI dashboards without additional query overhead on the source database.
- Developed automated ETL scripts in Python to scrape, clean, and normalize external datasets, replacing a manual data entry workflow and saving several hours of engineering time per week.

Trivia Softwares Mumbai, India
Software Developer Intern *Aug 2021 – Sep 2021*

- Built a Django REST API with secure user authentication and email verification, improving onboarding security and reducing manual account management overhead.

Projects

Traceline - Pipeline Anomaly Detection | *Python, AWS, React* | **1st Place — TidalHack @ TAMU**

- Designed an ingestion and alignment pipeline processing 15+ years of inspection records for RCP Inc., correlating multi-run scan data to track corrosion growth and predict structural failure risks.
- Engineered a relative-matching algorithm using fixed hardware anchors to correct odometer drift across inspection runs, ensuring cross-run data consistency for accurate anomaly co-registration.
- Built a React-based 3D visualization dashboard enabling field engineers to interactively track anomaly evolution across pipeline segments over time.

Data Centric Hackathon | *Python, PyTorch, 3LC, AWS* | **3rd Place - 3LC × AWS Hackathon @ UTD**

- Optimized a ResNet-18 classifier under strict constraints - no architecture changes, no external data, no pretrained weights - pushing validation accuracy to 94% purely through data quality improvements.
- Built an iterative data quality pipeline using 3LC Dashboard to relabel low-confidence samples, rebalance class distributions, and tune per-sample loss weights from embedding cluster analysis across training runs.